

Product Requirements Document (PRD)

Sea Lion Education

Offline-first educational AI platform for Indonesian schools in outermost and underdeveloped regions

Version: 1.0 (MVP)

Status: Draft

Target release: End of 2026

Audience: Engineers, UI/UX designers, researchers, university stakeholders, and implementation partners

Executive summary

Sea Lion Education is a smart-client educational AI platform designed for Indonesian schools in outermost, underdeveloped, and resource-constrained regions where internet access is unavailable or unreliable. The platform runs on a local school device and can optionally connect to cloud services when internet access is available. **Elise**, short for E-Library Sea Lion, serves K–12, junior high, and senior high school students by enabling interactive learning through an instant messaging-style interface, with an initial focus on Indonesian and English language learning.

The MVP delivers a lightweight conversational learning experience powered by a Sea Lion Small Language Model (SLM) and a Retrieval-Augmented Generation (RAG) system built from Indonesian curriculum books. It supports three core learning modes: Ask, Learn, and Practice.

1. **Ask mode:** its simple conversation mode such as “Berikan contoh soal hukum Newton 2?”. The AI will work to answer the question.
2. **Learn mode:** Students engage in a guided conversation to understand a selected topic, such as “Saya ingin belajar bab gradien di matematika kelas 2.” The AI supports learning by asking follow-up questions, explaining concepts, and helping students achieve the intended learning outcomes.
3. **Practice mode:** Students practice for assessments such as daily quizzes, midterms, finals, or national exams in a specific subject. For example, a student may request, “Saya ingin menyiapkan diri untuk berlatih ulangan harian termodinamika.” The AI generates multiple-choice, short-answer, or essay questions, evaluates the responses, calculates a score, and provides improvement recommendations.

The primary objective of the MVP is to validate that a locally deployed AI education platform can run smoothly on limited hardware while delivering reliable, curriculum-grounded learning support for students. **Elise** is a smart client platform which means it can work online and offline.

Problem statement

Students in many Indonesian outermost, underdeveloped, and resource-constrained regions have limited access to reliable learning support, qualified teaching assistance, and AI-powered educational tools. Although curriculum books may be available, students often lack an interactive system that can help them ask questions, understand topics, and practice assessments when internet connectivity is unavailable or unstable.

The core problem is the absence of accessible, curriculum-grounded learning assistance that can operate locally on limited school hardware while still allowing optional cloud connectivity for content updates and service enhancement when internet access is available.

Elise addresses this gap by bringing Sea Lion Education directly into schools as a smart-client educational AI platform. The system runs on a local school device, stores curriculum knowledge

locally, supports offline operation, and can optionally connect to cloud services to update learning materials or improve the knowledge base when connectivity is available.

Vision

To make curriculum-grounded AI learning accessible to every Indonesian student by delivering Elise as a smart-client educational platform that works offline on local school devices, supports Ask, Learn, and Practice interactions, and can connect to cloud services when available without making internet access a dependency.

MVP goal

Primary goal

Build Elise as a smart-client educational AI platform that enables students to ask questions, learn topics, and practice assessments using Indonesian curriculum books through a simple conversational interface. The MVP must run on a local school device, work without internet dependency, and optionally connect to cloud services when connectivity is available.

Success criteria

The MVP is considered successful if:

1. The platform runs reliably on a **local school device or cloud-based server**.
2. Students can complete **Ask, Learn, and Practice** workflows.
3. The AI remains grounded in the curriculum knowledge base.
4. Optional cloud connectivity can **support content updates** and **buy service enhancement** without becoming required dependency.

Goals and non-goals

Goals

1. Smart-client operation with offline-first capability
2. Local school device and local network accessibility
3. Curriculum-grounded AI responses
4. Grade-based learning personalization
5. Practice generation and grading
6. Teacher monitoring dashboard
7. Knowledge base management
8. Optional cloud connectivity for curriculum updates and knowledge base enhancement

Non-goals

1. Personalized learning roadmaps
2. Internet-dependent learning features
3. Mobile application
4. Mandatory online synchronization during learning sessions
5. Collaborative classrooms
6. Assignment management

6. Payment or subscription systems
7. Advanced analytics
8. Multi-school deployment through a shared central server

Target users

Primary user: Student

Profile

- Kindergarten to senior high school
- Limited technical literacy
- Using the system through a school computer or BYOD (Bring your own device)
- Requires simple and intuitive interactions

Primary goals

- Understand school subjects
- Interactive learning guide
- Practice and improve learning outcomes

Notes: student model will be available in version 1.0 and version 2.0.

- Version 1.0 – Standard mode
- Version 2.0 – Extended mode (such as subscription to additional content)

Secondary user: Teacher (Admin)

Profile

- School teacher
- Administrative access
- Responsible for monitoring students
- Can manage the knowledge base

Primary goals

- Review student progress
- Push additional content to the system

Notes: Teacher mode will be available in version 3.0.

Product overview

Elise, short for E-Library Sea Lion, is the student-facing product within Sea Lion Education. It provides a smart-client conversational learning experience that runs on a local school device, supports offline use, and can optionally connect to cloud services when internet access is available. The platform includes three core learning modes: Ask, Learn, and Practice.

Ask mode

Ask mode allows students to ask direct questions about a subject or topic using a simple instant messaging-style interface.

The AI retrieves relevant content from the curriculum knowledge base and responds with grounded explanations, examples, or clarifications. If the answer cannot be supported by available materials, the AI clearly indicates that the information is not available from the current knowledge base.

Learn mode

Learn mode provides guided learning conversations for students who want to understand a selected topic step by step.

Students can state what they want to learn, such as a subject, or topic. The AI then explains the topic, asks follow-up questions, checks understanding, and helps the student progress toward the intended learning outcomes.

Practice mode

Practice mode helps students prepare for assessments such as daily quizzes, midterms, finals, or national exams by generating exercises from the curriculum knowledge base just like NotebookLM user experience when generating Quiz.

- Topic
- Question type
- Number of questions

Supported question types:

- Multiple choice
- Short answer

- Essay

The AI supports multiple-choice, short-answer, and essay questions. It evaluates student responses, explains correctness, calculates a final score, and provides recommendations for improvement based on topic mastery.

User flow

Student onboarding

1. Register account
2. Create username and password
3. Select current grade
4. Enter platform

The selected grade determines the default curriculum knowledge available in Learn mode.

Students can update their grade later by visiting

Learn flow (Normal Conversation)

1. Open Learn
2. Ask educational question
3. AI retrieves **grade-specific** curriculum content
4. AI provides conversational explanation
5. Conversation is saved

Learn flow (Guided-Learning)

1. Open Learn
2. Select guided-learning mode
3. Select course
4. Define topic
5. AI retrieves **grade-specific** curriculum content
6. AI will ask educational question to train our ability on a certain topic
7. AI provides conversational explanation
8. Conversation is saved

1.

Practice flow

1. Open Practice
2. Select course
3. Define specific topic

4. Select question type
5. Select number of questions
6. Answer questions
7. Receive grading
8. View score and mastery summary
9. Session is saved

Functional requirements

Authentication

Student

- Register
- Login
- Logout
- Change password

Teacher

- Login
- Logout
- Administrative access

Email verification is not required because the system operates entirely **offline**.

Grade management

Students must have a grade associated with their account.

The system uses this grade to determine available curriculum materials.

Teachers can update student grades if necessary.

Learn mode (Normal Conversation)

The system shall:

- Search all curriculum books for the student's grade
- Maintain conversation context within the session
- Provide educational explanations
- Adapt language complexity to the student's grade
- Store conversation history

Learn mode (Guided-Learning)

The system shall:

- Search all curriculum books for the student's grade
- Maintain conversation context within the session
- Train student ability via active conversation (ask -> explain -> loop)
- Adapt language complexity to the student's grade
- Store conversation history

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Practice mode

The system shall:

- Generate questions from selected material
- Support MCQ, short answer, and essay
- Automatically evaluate responses
- Explain correct and incorrect answers
- Calculate final scores
- Produce topic mastery summaries
- Save practice history

Conversation history

Conversation history shall be separated by mode.

Students can:

- View history
- Rename conversations

Conversations persist across logins.

Progress tracking

Students shall have a progress page displaying:

- Practice history
- Scores
- Topic mastery
- Recent activity

Teacher dashboard

Teachers shall be able to view:

- Student accounts
- Learn conversations
- Ask conversations
- Practice history
- Scores
- Weak topics
- Usage frequency
- Most used subjects
- Average practice scores
- Students needing attention

Teachers cannot assign exercises in the MVP.

Knowledge base management

Teachers can:

- Upload books
- Organize materials
- Trigger automatic processing

Knowledge organization:

- School level
- Grade
- Subject
- Chapter

Uploaded books become available immediately after processing.

AI behavior requirements

Retrieval grounding

All educational responses must be grounded in the curriculum knowledge base.

Hallucination handling

If sufficient information is unavailable, the AI must explicitly state that the answer cannot be determined from the available books.

Language adaptation

The AI shall adjust:

- Vocabulary
- Explanation complexity
- Sentence structure

based on the student's grade level.

Tutor personality

The AI should behave as a friendly and encouraging tutor rather than a formal lecturer.

Safety

The AI must refuse:

- Harmful requests
- Inappropriate content
- Dangerous instructions
- Non-educational unsafe interactions

Offline-first requirements

Deployment

- One server per school
- Local network only
- No internet dependency during operation

Architecture assumptions

The system will run entirely within the school environment.

Student devices connect through the local network.

Each school has isolated data.

Performance assumptions

Expected concurrent users:

10–20

Streaming responses should be supported for a smoother user experience.

Data management

Stored data

- User accounts
- Grade information
- Conversation history
- Practice history
- Scores
- Knowledge base metadata

Retention

Student data remains available across sessions.

Teachers can permanently delete student data.

User interface principles

The interface should prioritize simplicity and accessibility.

Key principles:

- Large navigation elements
- Clear educational language
- Minimal technical terminology
- Mobile-responsive layout
- Dark mode support

The design should resemble a modern educational platform rather than a developer-oriented AI chat interface.

Error handling

AI failure

- Retry request
- Display friendly error message

Unsupported knowledge

Display:

“I don’t know based on the available books.”

Excessive input

Limit message length before submission.

Risks and assumptions

Risks

- Limited server hardware
- Response quality limitations
- Curriculum coverage gaps
- Knowledge base processing errors

Assumptions

- Curriculum books are available
- Sea Lion SLM can operate locally
- Schools have functioning local networks
- Teachers can perform basic administrative tasks

Future roadmap (out of scope)

Future versions may include:

- Personalized learning roadmaps
- Adaptive difficulty
- Learning recommendations
- Teacher assignments
- Multi-school administration
- Learning analytics
- Speech interaction
- Image understanding
- Offline content synchronization
- Student performance prediction

MVP acceptance criteria

The MVP is complete when:

1. Students can register and log in.
2. Students can select a grade.
3. Learn mode retrieves grade-specific knowledge.
4. Ask mode retrieves selected book knowledge.
5. Practice mode generates and grades exercises.
6. Scores and mastery summaries are stored.
7. Conversation history persists.
8. Teachers can monitor student activity.
9. Teachers can upload curriculum books.
10. The entire system operates successfully without internet connectivity on a local school server.